

Introduction to Trigonometry (Set B) — MCQ Worksheet

Mathematics • Chapter 8 • 25 Questions, 1 mark each — Total: 25 marks

Instructions: Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.

Q1. If $\sin A = 4/5$, then $\cos A =$

- | | |
|-----------|-----------|
| (A) $3/5$ | (B) $4/5$ |
| (C) $5/3$ | (D) $5/4$ |

Q2. If $\tan A = 3/4$, $\sin A =$

- | | |
|-----------|-----------|
| (A) $3/5$ | (B) $4/5$ |
| (C) $5/3$ | (D) $5/4$ |

Q3. $\sin 60^\circ \times \cos 30^\circ =$

- | | |
|-----------|------------------|
| (A) $1/2$ | (B) $3/4$ |
| (C) 1 | (D) $\sqrt{3}/2$ |

Q4. Value of $2 \sin 30^\circ + 3 \cos 60^\circ$:

- | | |
|-----------|-----------|
| (A) $5/2$ | (B) $3/2$ |
| (C) 1 | (D) 2 |

Q5. If $\cos \theta = 12/13$, then $\sin \theta =$

- | | |
|------------|-------------|
| (A) $5/13$ | (B) $5/12$ |
| (C) $12/5$ | (D) $13/12$ |

Q6. $\operatorname{cosec} 30^\circ + \cot 45^\circ =$

- | | |
|---------|---------|
| (A) 3 | (B) 2 |
| (C) 1 | (D) 0 |

Q7. If $\tan \theta = 1$, $\sec \theta =$

- | | |
|------------------|---------|
| (A) $\sqrt{2}$ | (B) 1 |
| (C) $1/\sqrt{2}$ | (D) 2 |

Q8. Value of $\sin 45^\circ + \cos 45^\circ$:

- | | |
|----------------|------------------|
| (A) $\sqrt{2}$ | (B) $1/\sqrt{2}$ |
| (C) 1 | (D) 2 |

Q9. If $\sin (A + B) = 1$ and $\cos (A - B) = 1$, then $(A, B) =$

- | | |
|----------------------------|-------------------------------|
| (A) $(45^\circ, 45^\circ)$ | (B) $(60^\circ, 30^\circ)$ |
| (C) $(30^\circ, 60^\circ)$ | (D) Both A and B = 45° |

Q10. $(1 - \cos^2 \theta) =$

- | | |
|---------------------|---------------------|
| (A) $\sin^2 \theta$ | (B) $\tan^2 \theta$ |
| (C) $\cot^2 \theta$ | (D) $\sec^2 \theta$ |

Q11. $(\sec^2 \theta - 1) =$

- | | |
|---------------------|-------------------------------------|
| (A) $\sin^2 \theta$ | (B) $\tan^2 \theta$ |
| (C) $\cot^2 \theta$ | (D) $\operatorname{cosec}^2 \theta$ |

Q12. Value of $(\sin^2 30^\circ + \cos^2 30^\circ)$:

- | | |
|-----------|------------------|
| (A) 0 | (B) 1 |
| (C) $1/2$ | (D) $\sqrt{3}/2$ |

Q13. If $\cot A = 12/5$, then $\sin A =$

- | | |
|------------|-------------|
| (A) $5/13$ | (B) $12/13$ |
| (C) $13/5$ | (D) $13/12$ |

Q14. $\sin 90^\circ - \cos 90^\circ =$

- | | |
|----------|---------|
| (A) 1 | (B) 0 |
| (C) -1 | (D) 2 |

Q15. $2 \tan 30^\circ / (1 - \tan^2 30^\circ) =$

- (A) $\sin 60^\circ$
(C) $\tan 60^\circ$

- (B) $\cos 60^\circ$
(D) $\tan 30^\circ$

Q16. If $\tan \theta + \cot \theta = 5$, then $\tan^2 \theta + \cot^2 \theta =$

- (A) 23
(C) 27

- (B) 25
(D) 21

Q17. $(1 + \tan^2 A)/(1 + \cot^2 A) =$

- (A) $\sec^2 A$
(C) $\cot^2 A$

- (B) -1
(D) $\tan^2 A$

Q18. If $\sec \theta + \tan \theta = p$, then $\sec \theta - \tan \theta =$

- (A) p
(C) p^2

- (B) $1/p$
(D) $-p$

Q19. Value of $(\sin 18^\circ / \cos 72^\circ)$:

- (A) 1
(C) -1

- (B) 0
(D) 2

Q20. If $3 \sin \theta = 4 \cos \theta$, then $3 \sin \theta + 5 \cos \theta / 3 \sin \theta - 5 \cos \theta =$

- (A) -9
(C) -7

- (B) 9
(D) 7

Q21. Prove: $(\operatorname{cosec} \theta - \cot \theta)^2 = (1 - \cos \theta)/(1 + \cos \theta)$. Identity is:

- (A) Correct (LHS = RHS always)
(C) Conditional

- (B) Incorrect
(D) Undefined

Q22. If $\sin \theta + \cos \theta = \sqrt{2}$, then $\tan \theta + \cot \theta =$

- (A) 1
(C) $\sqrt{2}$

- (B) 2
(D) 0

Q23. If $7 \sin^2 \theta + 3 \cos^2 \theta = 4$, then $\tan \theta =$

- (A) $1/\sqrt{3}$
(C) 1

- (B) $\sqrt{3}$
(D) 0

Q24. If $\sin \theta = x$ and $\cos \theta = 1/x$, find x .

- (A) No real x exists
(C) -1

- (B) 1
(D) 0

Q25. $(\sec A - \tan A)(\sec A + \tan A) =$

- (A) 0
(C) $\sec^2 A$

- (B) 1
(D) $\tan^2 A$